

Supplementary Material for Sliding-Window Reordering with Overlap Averaging: A Simple Time-Domain Augmentation for Multivariate Forecasting

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This document provides supplementary details for the main paper. It contains: (A) augmentation baselines and hyperparameters, including TPS hyperparameter selection, (B) complementary long-term and short-term forecasting results (per-dataset, per-length, and with standard deviations), (C) extended ablation studies (component-wise with standard deviations, augmentation size, augmentation ratio), and (D) dataset statistics and experimental setup for time series classification.

Section, table, and figure numbering inside this document is independent of the main paper. Full per-dataset and per-length Excel tables are additionally available in the code repository: <https://github.com/jafarbakhshaliyev/TPS/blob/main/results/results.xlsx>.

1 Augmentation Baselines and Hyperparameters

Table 1 summarizes the hyperparameters used for all augmentation methods evaluated in this study, including our proposed method TPS. Most augmentation methods require only a small number of hyperparameters, and their recommended values are provided in the original papers. Nevertheless, we re-evaluated these settings to verify their effectiveness within our experimental setup.

For wDBA and MBB, we reproduce the methods based on the descriptions provided in their original papers [3, 7]. For RobustTAD, we follow the implementation details from [8]: amplitude-based augmentation replaces segments with values sampled from a Gaussian distribution, while phase-based augmentation perturbs phase values using Gaussian noise. FreqAdd and FreqPool are implemented following [5, 12], where the former modifies the frequency components and the latter compresses the spectrum. For Upsample, we reproduce the method by selecting a consecutive segment and stretching it to match the length of the original time series [9]. For STAug, Freq-Mask/Mix, Wave-Mask/Mix, and Dominant Shuffle, we directly use the official implementations from their released codebases [1, 4, 11, 13].

Table 1: Hyperparameters for the time series forecasting augmentation methods.

Method	Hyperparameters
wDBA [7]	weighting, DTW constraints
MBB [3]	block size, STL period
RobustTAD-m/p [8]	perturbation rate, #segments, segment length
FreqAdd [12]	perturbation rate
FreqPool [5]	pool size
Upsample [9]	subsequence length rate
STAug [11]	mixup rate
FreqMask [4]	masking rate
FreqMix [4]	mixing rate
Wave-Mask/Mix [1]	wavelet type, decomposition level, sampling rate
Dominant Shuffle [13]	shuffle rate
TPS (Ours)	patch length (p), stride (s), shuffle rate (α)

1.1 TPS Hyperparameter Selection

We select (p, s, α) using a validation-based search over a predefined set of candidate values:

- Patch length:

$$p \in \{16, 32, 48, 64, 72, 96, 120, 168, 192, 200, 220, 240, 280, 300, 340, 380, 400, 420, 440, 560, 700\}.$$

- Stride: $s \in \{1, 2, 5, 8, 12, 16, 24, 32, 36, 96\}$.
- Shuffle rate: $\alpha \in \{0.2, 0.5, 0.7, 0.8, 0.9, 1.0\}$.

To limit computational cost, we do not evaluate the full Cartesian product. Instead, we test a fixed set of approximately 20 candidate (p, s, α) combinations and select the one that minimizes validation MSE. The selected configuration is then evaluated once on the test set.

2 Complementary Forecasting Results

2.1 Long-Term Forecasting on the Common Seven-Dataset Subset

Table 2 reports the same long-term forecasting results as Table 2 in the main paper, but averaged over the common subset of seven

datasets (excluding ECL and Traffic). This additional view is necessary because STAug cannot be evaluated on ECL and Traffic due to its high GPU memory requirements, consistent with the original STAug paper [11].

TPS achieves substantial improvements of **2.98%**, **5.83%**, **3.16%**, **2.26%**, and **10.90%** for TSMixer, DLinear, PatchTST, TiDE, and LightTS, respectively, while also obtaining a high number of wins across all models. We also observe that STAug underperforms several other augmentation baselines on this common subset.

2.2 Long-Term Forecasting with Standard Deviations

Tables 3 and 4 report long-term forecasting results on eight benchmark datasets (excluding ILLI). We report mean MSE/MAE $\pm$ std, where the mean and standard deviation are first computed over five runs for each prediction length and then averaged over {96, 192, 336, 720}.

TPS achieves the best performance in most settings, typically ranking first and occasionally second. The strongest competing methods vary by dataset, but are most often Freq-Mask/Mix, Wave-Mask/Mix, Dominant Shuffle, or Upsample. In addition to improved accuracy, TPS often reduces variability relative to training without augmentation, indicating that it can introduce useful diversity while keeping perturbations controlled.

Full results for each individual prediction length are also available in the supplementary Excel file released with the code: <https://github.com/jafarbakshaliyev/TPS/blob/main/results/results.xlsx>.

2.3 Short-Term Forecasting Per Prediction Length

Table 5 reports the short-term forecasting results (MSE and MAE) on PeMS-{03, 04, 07, 08} for prediction lengths {12, 24, 36, 48}, averaged over five runs. TPS achieves the best performance in most cases, except at the 48-step prediction length where it generally ranks second. The strongest competing methods are typically STAug, FreqPool, or Wave-Mask/Mix.

Experiments in this setting are conducted using PatchTST, which serves as a strong representative backbone on the PeMS benchmarks. This choice keeps the short-term evaluation tractable while still providing a meaningful comparison across augmentation methods on large, high-dimensional traffic datasets.

3 Further Ablation Studies

Full Component-wise Analysis. Table 6 presents the complete component-wise ablation results on the ETT datasets, including MSE and MAE with their corresponding standard deviations. For each prediction length in {96, 192, 336, 720}, results are computed over five runs and then averaged across the four prediction lengths. The MAE results follow the same trends as the MSE results, further supporting the consistency of each component’s contribution.

First, we remove the variance-based ordering used to prioritize patches for shuffling. The results indicate that this component generally provides a modest improvement, although its effect naturally vanishes when the shuffle rate is set to 1.0.

Second, we replace overlapping patches with non-overlapping ones, which leads to a substantial degradation in performance. This

finding supports our design choice in TPS: overlapping patches are important for preserving local temporal structure and reducing discontinuities at patch boundaries.

Third, we examine the role of data-label coherence in the augmentation pipeline. In the default design, augmentation is applied jointly to the input and forecast horizon. When augmentation is applied only to the input, performance deteriorates significantly, consistent with prior observations [4, 10] on the importance of maintaining input-target alignment.

Finally, we evaluate a frequency-domain variant in which the same patch-based operations are applied after transforming the signal using the Fast Fourier Transform (FFT). This variant also degrades performance, indicating that TPS is more effective in the time domain.

Impact of Varying Augmentation Sizes. Figure 1 reports an ablation study with varying augmentation sizes (1, 2, 3, 4, and 5) using PatchTST on ETTh1 and ETTh2 with prediction length 96. An augmentation size of 2 means that the augmented sample set is doubled by applying the augmentation method twice. This analysis examines whether increasing augmentation intensity continues to improve performance or instead introduces overly strong perturbations.

The results show that FreqMix benefits from larger augmentation sizes on both datasets, while FreqMask improves only on ETTh2 when applied twice. In contrast, most other methods degrade as augmentation size increases. Notably, TPS on ETTh1 exhibits only minor performance variation up to augmentation size 4, indicating stable behavior under stronger augmentation. This contrasts with methods such as Upsample and FreqMask, which are more sensitive to augmentation size. Dominant Shuffle and FreqMix also remain relatively stable across different augmentation sizes. On ETTh2, TPS shows some degradation at larger augmentation sizes, suggesting that overly strong perturbations can become harmful in this setting. Nevertheless, across other models and prediction lengths on ETTh2, TPS is generally stable under different augmentation sizes.

Impact of Varying Augmentation Ratios. We also conduct experiments with different augmentation ratios (0.1, 0.3, 0.5, 0.7, and 1.0) using PatchTST on ETTh1 and ETTh2 with prediction length 96, as shown in Figure 2. The evaluated methods include FreqMask, FreqMix, Upsample, Dominant Shuffle, and TPS. Here, the augmentation ratio denotes the proportion of augmented samples included in each training batch, so lower ratios correspond to fewer augmented samples. TPS achieves the lowest MSE on both datasets when using the full augmentation ratio (1.0). Moreover, even with only 10% augmented samples (ratio 0.1), TPS outperforms all competing augmentation methods at their respective ratios.

4 TPS for Time Series Classification

Datasets and Experimental Setups. We use the widely adopted UCR [6] and UEA [2] repositories to evaluate TPS on univariate and multivariate time series classification tasks.

For univariate classification, we use the UCR archive, which contains datasets with a single time-dependent variable (i.e., one channel). The archive covers diverse categories, including Device, ECG, Image, Motion, Sensor, Spectrograph, Simulated, and others [6].

Table 2: Long-term forecasting performance (MSE/MAE) averaged over seven datasets and four prediction lengths. #Wins is counted over 28 settings (7 datasets $\times$ 4 prediction lengths). Best results are highlighted in bold and second-best results are underlined. (ECL and Traffic are excluded to match the common subset used by all methods.)

Method	TSMixer		DLinear		PatchTST		TiDE		LightTS	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
None	0.506	0.433	0.619	0.484	0.519	0.437	0.535	0.443	0.744	0.532
wDBA [7]	0.507	<u>0.432</u>	0.603	0.477	0.508	0.434	0.551	0.454	0.730	0.525
MBB [3]	0.514	0.435	0.611	0.480	0.507	<u>0.432</u>	0.550	0.451	0.746	0.533
RobustTAD-m/p [8]	0.508	0.432	0.610	0.480	0.515	0.445	0.538	0.444	0.748	0.532
FreqAdd [12]	<u>0.504</u>	0.433	0.628	0.489	0.518	0.438	0.533	0.443	0.739	0.530
FreqPool [5]	0.520	0.439	0.593	0.475	0.529	0.442	0.551	0.452	0.709	0.515
Upsample [9]	0.520	0.437	<u>0.583</u>	<u>0.465</u>	0.519	0.437	0.567	0.452	<u>0.688</u>	<u>0.504</u>
STAug [11]	0.540	0.451	0.733	0.538	0.548	0.453	0.630	0.497	0.855	0.583
Freq-Mask/Mix [4]	0.512	0.435	0.608	0.475	0.512	0.438	0.546	0.447	0.729	0.524
Wave-Mask/Mix [1]	0.509	0.433	0.615	0.481	<u>0.507</u>	0.436	<u>0.531</u>	<u>0.441</u>	0.739	0.527
Dominant Shuffle [13]	0.517	0.437	0.615	0.477	0.521	0.442	0.548	0.446	0.725	0.518
TPS (Ours)	0.489	0.424	0.549	0.448	0.491	0.424	0.519	0.434	0.613	0.478
#Wins (out of 28)	20	22	28	28	22	24	27	25	27	27
Improvement	2.98%	1.85%	5.83%	3.66%	3.16%	1.89%	2.26%	1.59%	10.90%	5.16%

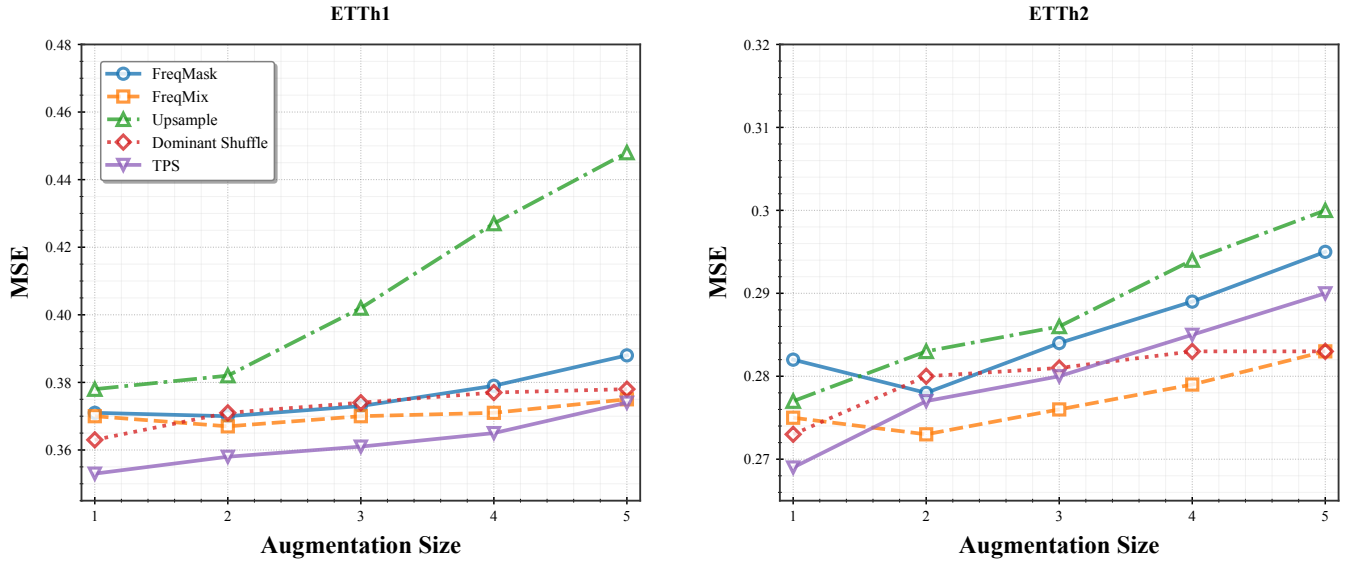


Figure 1: Impact of varying augmentation sizes (1–5) on forecasting performance using the PatchTST model with a prediction length of 96 on the ETTh1 and ETTh2 datasets. The results demonstrate how different augmentation methods respond to increasing augmentation intensity, highlighting stability or degradation in performance.

For our evaluation, we select 30 datasets from the 128 datasets in UCR, ensuring coverage across multiple categories. These datasets vary in training and test sizes, sequence lengths, and numbers of classes, making the evaluation broad and representative. Table 7 summarizes the selected UCR datasets.

For multivariate classification, we use the UEA archive [2], which contains datasets with multiple input channels. We select 10 datasets from the 30 datasets in UEA, covering diverse application domains and a broad range of input dimensionalities, sequence lengths, and class counts. Table 8 summarizes the selected UEA datasets.

For both settings, we split the original training data into 80% training and 20% validation sets. Hyperparameters are selected based on validation accuracy. When multiple configurations achieve the same validation accuracy, one is randomly chosen for final training on the full training set and evaluation on the provided test set.

Table 3: Long-term forecasting results averaged over prediction lengths {96, 192, 336, 720} on ETT-{h1,h2,m1,m2} for all five backbone models. For each prediction length, we report mean $\pm$ std over 5 runs, then average these statistics over the four lengths. Best results per dataset and model are highlighted in bold, and second-best results are underlined.

	Method	ETTh1		ETTh2		ETTm1		ETTm2	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TSMixer	None	0.413 ± 0.0023	0.434 ± 0.0020	0.342 ± 0.0023	0.394 ± 0.0017	0.351 ± 0.0010	0.378 ± 0.0008	0.262 ± 0.0017	0.320 ± 0.0014
	wDBA [7]	0.418 ± 0.0028	0.437 ± 0.0026	0.344 ± 0.0023	0.394 ± 0.0019	0.352 ± 0.0009	0.378 ± 0.0008	0.261 ± 0.0011	0.319 ± 0.0011
	MBB [3]	0.412 ± 0.0026	0.437 ± 0.0023	0.345 ± 0.0046	0.395 ± 0.0034	0.353 ± 0.0014	0.379 ± 0.0013	0.264 ± 0.0017	0.321 ± 0.0011
	RobustTAD-m/p [8]	0.416 ± 0.0018	0.434 ± 0.0016	0.343 ± 0.0032	0.394 ± 0.0023	0.350 ± 0.0008	0.377 ± 0.0008	0.262 ± 0.0037	0.319 ± 0.0021
	FreqAdd [12]	0.413 ± 0.0026	0.432 ± 0.0024	0.343 ± 0.0025	0.393 ± 0.0016	0.351 ± 0.0010	0.377 ± 0.0008	0.263 ± 0.0062	0.320 ± 0.0035
	FreqPool [5]	0.431 ± 0.0055	0.446 ± 0.0043	0.351 ± 0.0031	0.399 ± 0.0020	0.359 ± 0.0010	0.382 ± 0.0010	0.264 ± 0.0015	0.322 ± 0.0013
	Upsample [9]	0.416 ± 0.0023	0.434 ± 0.0023	0.343 ± 0.0019	0.392 ± 0.0014	0.378 ± 0.0013	0.396 ± 0.0009	0.263 ± 0.0016	0.320 ± 0.0010
	STAug [11]	0.411 ± 0.0020	0.432 ± 0.0019	0.393 ± 0.0067	0.427 ± 0.0032	0.352 ± 0.0017	0.377 ± 0.0015	0.263 ± 0.0055	0.373 ± 0.0021
	Freq-Mask/Mix [4]	0.416 ± 0.0035	0.434 ± 0.0027	0.344 ± 0.0019	0.394 ± 0.0015	0.348 ± 0.0010	0.376 ± 0.0006	0.259 ± 0.0030	0.319 ± 0.0018
	Wave-Mask/Mix [1]	0.410 ± 0.0021	0.431 ± 0.0021	0.342 ± 0.0022	0.393 ± 0.0016	0.352 ± 0.0013	0.379 ± 0.0009	0.261 ± 0.0015	0.320 ± 0.0009
	Dominant Shuffle [13]	0.416 ± 0.0019	0.434 ± 0.0023	0.345 ± 0.0027	0.395 ± 0.0020	0.353 ± 0.0012	0.380 ± 0.0009	0.261 ± 0.0017	0.320 ± 0.0007
	TPS (Ours)	0.402 ± 0.0018	0.425 ± 0.0019	0.339 ± 0.0021	0.390 ± 0.0015	0.346 ± 0.0010	0.373 ± 0.0007	0.257 ± 0.0016	0.317 ± 0.0008
DLinear	None	0.438 ± 0.0194	0.449 ± 0.0162	0.464 ± 0.0099	0.462 ± 0.0053	0.361 ± 0.0010	0.383 ± 0.0015	0.276 ± 0.0099	0.339 ± 0.0093
	wDBA	0.432 ± 0.0126	0.443 ± 0.0126	0.434 ± 0.0383	0.444 ± 0.0175	0.360 ± 0.0011	0.381 ± 0.0016	0.270 ± 0.0050	0.334 ± 0.0055
	MBB	0.429 ± 0.0114	0.444 ± 0.0111	0.449 ± 0.0324	0.454 ± 0.0152	0.360 ± 0.0009	0.382 ± 0.0015	0.280 ± 0.0080	0.343 ± 0.0072
	RobustTAD-m/p	0.433 ± 0.0155	0.444 ± 0.0129	0.432 ± 0.0295	0.447 ± 0.0151	0.361 ± 0.0012	0.383 ± 0.0017	0.280 ± 0.0088	0.342 ± 0.0071
	FreqAdd	0.432 ± 0.0128	0.446 ± 0.0120	0.471 ± 0.0195	0.466 ± 0.0090	0.364 ± 0.0009	0.386 ± 0.0012	0.285 ± 0.0097	0.347 ± 0.0085
	FreqPool	0.450 ± 0.0231	0.455 ± 0.0166	0.422 ± 0.0177	0.439 ± 0.0104	0.380 ± 0.0008	0.399 ± 0.0011	0.273 ± 0.0111	0.335 ± 0.0075
	Upsample	0.436 ± 0.0059	0.445 ± 0.0055	0.391 ± 0.0178	0.423 ± 0.0098	0.387 ± 0.0007	0.406 ± 0.0015	0.265 ± 0.0048	0.328 ± 0.0048
	STAug	0.442 ± 0.0345	0.450 ± 0.0226	1.011 ± 0.1382	0.678 ± 0.0543	0.359 ± 0.0016	0.383 ± 0.0023	0.442 ± 0.0212	0.416 ± 0.0131
	Freq-Mask/Mix	0.422 ± 0.0063	0.436 ± 0.0066	0.422 ± 0.0246	0.440 ± 0.0133	0.360 ± 0.0009	0.383 ± 0.0013	0.271 ± 0.0037	0.336 ± 0.0044
	Wave-Mask/Mix	0.426 ± 0.0058	0.441 ± 0.0056	0.454 ± 0.0228	0.456 ± 0.0118	0.361 ± 0.0014	0.383 ± 0.0019	0.275 ± 0.0108	0.338 ± 0.0096
	Dominant Shuffle	0.420 ± 0.0037	0.434 ± 0.0044	0.409 ± 0.0104	0.435 ± 0.0066	0.360 ± 0.0014	0.383 ± 0.0015	0.271 ± 0.0089	0.335 ± 0.0095
	TPS (Ours)	0.410 ± 0.0036	0.425 ± 0.0031	0.369 ± 0.0056	0.408 ± 0.0039	0.354 ± 0.0006	0.377 ± 0.0006	0.261 ± 0.0018	0.324 ± 0.0027
PatchTST	None	0.414 ± 0.0026	0.428 ± 0.0023	0.331 ± 0.0016	0.380 ± 0.0020	0.354 ± 0.0030	0.383 ± 0.0019	0.258 ± 0.0018	0.316 ± 0.0013
	wDBA	0.417 ± 0.0047	0.431 ± 0.0035	0.335 ± 0.0009	0.382 ± 0.0011	0.352 ± 0.0019	0.381 ± 0.0010	0.261 ± 0.0011	0.318 ± 0.0010
	MBB	0.422 ± 0.0057	0.433 ± 0.0033	0.332 ± 0.0011	0.381 ± 0.0014	0.353 ± 0.0019	0.383 ± 0.0008	0.258 ± 0.0004	0.316 ± 0.0007
	RobustTAD-m/p	0.423 ± 0.0116	0.434 ± 0.0058	0.331 ± 0.0009	0.380 ± 0.0059	0.354 ± 0.0025	0.383 ± 0.0018	0.258 ± 0.0020	0.317 ± 0.0016
	FreqAdd	0.413 ± 0.0022	0.429 ± 0.0018	0.333 ± 0.0013	0.381 ± 0.0019	0.355 ± 0.0041	0.384 ± 0.0014	0.261 ± 0.0022	0.317 ± 0.0016
	FreqPool	0.423 ± 0.0039	0.436 ± 0.0029	0.333 ± 0.0012	0.381 ± 0.0008	0.351 ± 0.0024	0.381 ± 0.0015	0.256 ± 0.0018	0.315 ± 0.0008
	Upsample	0.419 ± 0.0047	0.433 ± 0.0029	0.335 ± 0.0010	0.382 ± 0.0011	0.356 ± 0.0037	0.387 ± 0.0021	0.258 ± 0.0013	0.316 ± 0.0006
	STAug	0.419 ± 0.0074	0.429 ± 0.0033	0.416 ± 0.0115	0.428 ± 0.0047	0.348 ± 0.0025	0.379 ± 0.0016	0.275 ± 0.0036	0.325 ± 0.0017
	Freq-Mask/Mix	0.412 ± 0.0024	0.427 ± 0.0023	0.329 ± 0.0012	0.380 ± 0.0009	0.351 ± 0.0019	0.382 ± 0.0011	0.261 ± 0.0038	0.318 ± 0.0025
	Wave-Mask/Mix	0.412 ± 0.0029	0.428 ± 0.0021	0.330 ± 0.0017	0.379 ± 0.0018	0.352 ± 0.0027	0.381 ± 0.0018	0.258 ± 0.0012	0.316 ± 0.0012
	Dominant Shuffle	0.408 ± 0.0015	0.424 ± 0.0010	0.329 ± 0.0009	0.382 ± 0.0012	0.354 ± 0.0022	0.383 ± 0.0017	0.258 ± 0.0014	0.316 ± 0.0013
	TPS (Ours)	0.401 ± 0.0021	0.419 ± 0.0021	0.326 ± 0.0013	0.378 ± 0.0008	0.345 ± 0.0019	0.377 ± 0.0009	0.256 ± 0.0013	0.315 ± 0.0009
TiDE	None	0.417 ± 0.0012	0.432 ± 0.0011	0.316 ± 0.0018	0.375 ± 0.0011	0.359 ± 0.0036	0.381 ± 0.0032	0.250 ± 0.0009	0.313 ± 0.0005
	wDBA	0.563 ± 0.0055	0.518 ± 0.0023	0.321 ± 0.0009	0.378 ± 0.0006	0.360 ± 0.0027	0.382 ± 0.0025	0.250 ± 0.0006	0.313 ± 0.0009
	MBB	0.556 ± 0.0027	0.513 ± 0.0011	0.311 ± 0.0006	0.371 ± 0.0004	0.358 ± 0.0025	0.381 ± 0.0020	0.250 ± 0.0003	0.312 ± 0.0004
	RobustTAD-m/p	0.418 ± 0.0019	0.433 ± 0.0012	0.316 ± 0.0008	0.375 ± 0.0005	0.357 ± 0.0008	0.380 ± 0.0011	0.251 ± 0.0009	0.313 ± 0.0005
	FreqAdd	0.406 ± 0.0007	0.428 ± 0.0006	0.312 ± 0.0007	0.371 ± 0.0005	0.361 ± 0.0022	0.383 ± 0.0018	0.252 ± 0.0017	0.315 ± 0.0009
	FreqPool	0.431 ± 0.0032	0.442 ± 0.0026	0.322 ± 0.0010	0.380 ± 0.0006	0.377 ± 0.0021	0.395 ± 0.0016	0.253 ± 0.0009	0.316 ± 0.0004
	Upsample	0.423 ± 0.0008	0.439 ± 0.0006	0.328 ± 0.0043	0.381 ± 0.0023	0.390 ± 0.0042	0.402 ± 0.0023	0.252 ± 0.0005	0.314 ± 0.0003
	STAug	0.533 ± 0.0038	0.514 ± 0.0018	0.571 ± 0.0751	0.521 ± 0.0382	0.356 ± 0.0016	0.379 ± 0.0017	0.421 ± 0.0099	0.396 ± 0.0037
	Freq-Mask/Mix	0.421 ± 0.0017	0.436 ± 0.0011	0.317 ± 0.0008	0.377 ± 0.0007	0.355 ± 0.0011	0.379 ± 0.0009	0.253 ± 0.0012	0.317 ± 0.0007
	Wave-Mask/Mix	0.402 ± 0.0005	0.426 ± 0.0006	0.312 ± 0.0009	0.372 ± 0.0007	0.357 ± 0.0014	0.380 ± 0.0011	0.249 ± 0.0003	0.312 ± 0.0005
	Dominant Shuffle	0.412 ± 0.0010	0.429 ± 0.0007	0.312 ± 0.0010	0.374 ± 0.0005	0.353 ± 0.0020	0.379 ± 0.0018	0.255 ± 0.0007	0.317 ± 0.0006
	TPS (Ours)	0.387 ± 0.0010	0.415 ± 0.0008	0.308 ± 0.0005	0.370 ± 0.0004	0.347 ± 0.0019	0.373 ± 0.0017	0.248 ± 0.0003	0.311 ± 0.0003
LightTS	None	0.462 ± 0.0051	0.473 ± 0.0030	0.611 ± 0.0152	0.540 ± 0.0089	0.380 ± 0.0025	0.401 ± 0.0021	0.301 ± 0.0052	0.363 ± 0.0050
	wDBA	0.462 ± 0.0031	0.473 ± 0.0025	0.588 ± 0.0115	0.529 ± 0.0059	0.381 ± 0.0051	0.401 ± 0.0035	0.290 ± 0.0050	0.352 ± 0.0042
	MBB	0.454 ± 0.0026	0.469 ± 0.0023	0.612 ± 0.0157	0.541 ± 0.0071	0.382 ± 0.0030	0.404 ± 0.0027	0.306 ± 0.0081	0.367 ± 0.0076
	RobustTAD-m/p	0.462 ± 0.0023	0.473 ± 0.0015	0.597 ± 0.0127	0.535 ± 0.0074	0.382 ± 0.0032	0.402 ± 0.0029	0.299 ± 0.0021	0.360 ± 0.0025
	FreqAdd	0.454 ± 0.0032	0.468 ± 0.0031	0.615 ± 0.0129	0.541 ± 0.0061	0.378 ± 0.0028	0.400 ± 0.0025	0.300 ± 0.0099	0.361 ± 0.0091
	FreqPool	0.482 ± 0.0057	0.487 ± 0.0040	0.543 ± 0.0155	0.516 ± 0.0089	0.392 ± 0.0027	0.410 ± 0.0027	0.297 ± 0.0060	0.358 ± 0.0037
	Upsample	0.468 ± 0.0030	0.478 ± 0.0020	0.446 ± 0.0066	0.464 ± 0.0034	0.402 ± 0.0054	0.415 ± 0.0038	0.283 ± 0.0018	0.347 ± 0.0027
	STAug	0.458 ± 0.0034	0.470 ± 0.0026	1.248 ± 0.0336	0.782 ± 0.0119	0.363 ± 0.0014	0.388 ± 0.0014	0.546 ± 0.0165	0.486 ± 0.0083
	Freq-Mask/Mix	0.461 ± 0.0028	0.473 ± 0.0018	0.558 ± 0.0110	0.518 ± 0.0063	0.373 ± 0.0016	0.398 ± 0.0014	0.297 ± 0.0119	0.360 ± 0.0118
	Wave-Mask/Mix	0.448 ± 0.0031	0.466 ± 0.0021	0.611 ± 0.0086	0.539 ± 0.0047	0.377 ± 0.0018	0.400 ± 0.0024	0.299 ± 0.0075	0.360 ± 0.0074
Dominant Shuffle	0.449 ± 0.0036	0.464 ± 0.0028	0.511 ± 0.0101	0.496 ± 0.0058	0.372 ± 0.0038	0.398 ± 0.0031	0.290 ± 0.0035	0.353 ± 0.0043	
TPS (Ours)	0.431 ± 0.0007	0.452 ± 0.0005	0.418 ± 0.0031	0.447 ± 0.0019	0.356 ± 0.0015	0.382 ± 0.0015	0.277 ± 0.0027	0.342 ± 0.0029	

Table 4: Long-term forecasting results averaged over prediction lengths {96, 192, 336, 720} on Exchange, Weather, ECL, and Traffic for all five backbone models. For each prediction length, we report mean $\pm$ std over 5 runs, then average these statistics over the four lengths. Best results per dataset and model are highlighted in bold, and second-best results are underlined. STAUG is not reported on ECL and Traffic due to GPU memory limitations.

	Method	Exchange		Weather		ECL		Traffic	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
TSMixer	None	0.417 ± 0.0074	0.436 ± 0.0038	0.224 ± 0.0016	0.263 ± 0.0014	0.170 ± 0.0003	0.269 ± 0.0004	0.436 ± 0.0007	0.327 ± 0.0009
	wDBA [7]	0.406 ± 0.0091	0.430 ± 0.0050	0.222 ± 0.0006	0.261 ± 0.0006	0.173 ± 0.0002	0.275 ± 0.0002	0.437 ± 0.0010	0.326 ± 0.0013
	MBB [3]	0.408 ± 0.0074	0.431 ± 0.0038	0.224 ± 0.0006	0.263 ± 0.0006	0.171 ± 0.0001	0.271 ± 0.0002	0.438 ± 0.0069	0.325 ± 0.0077
	RobustTAD-m/p [8]	0.409 ± 0.0042	0.431 ± 0.0028	0.223 ± 0.0010	0.262 ± 0.0010	0.168 ± 0.0003	0.268 ± 0.0005	0.431 ± 0.0004	0.317 ± 0.0008
	FreqAdd [12]	0.407 ± 0.0071	0.436 ± 0.0035	0.226 ± 0.0007	0.265 ± 0.0006	0.168 ± 0.0002	0.268 ± 0.0002	0.432 ± 0.0006	0.319 ± 0.0009
	FreqPool [5]	0.418 ± 0.0032	0.434 ± 0.0021	0.224 ± 0.0009	0.264 ± 0.0011	0.176 ± 0.0004	0.277 ± 0.0004	0.442 ± 0.0009	0.325 ± 0.0005
	Upsample [9]	0.401 ± 0.0053	0.425 ± 0.0032	0.217 ± 0.0008	0.259 ± 0.0009	0.187 ± 0.0003	0.286 ± 0.0004	0.463 ± 0.0008	0.346 ± 0.0008
	STAug [11]	0.413 ± 0.0061	0.434 ± 0.0037	0.241 ± 0.0046	0.285 ± 0.0058	-	-	-	-
	Freq-Mask/Mix [4]	0.431 ± 0.0068	0.444 ± 0.0042	0.222 ± 0.0006	0.262 ± 0.0007	0.174 ± 0.0002	0.276 ± 0.0003	0.450 ± 0.0008	0.323 ± 0.0004
	Wave-Mask/Mix [1]	0.413 ± 0.0073	0.434 ± 0.0036	0.225 ± 0.0007	0.265 ± 0.0010	0.169 ± 0.0002	0.269 ± 0.0003	0.434 ± 0.0008	0.323 ± 0.0007
	Dominant Shuffle [13]	0.401 ± 0.0057	0.431 ± 0.0036	0.225 ± 0.0010	0.264 ± 0.0009	0.175 ± 0.0002	0.276 ± 0.0003	0.444 ± 0.0006	0.326 ± 0.0007
	TPS (Ours)	0.391 ± 0.0067	0.421 ± 0.0036	0.222 ± 0.0010	0.260 ± 0.0007	0.168 ± 0.0002	0.267 ± 0.0002	0.428 ± 0.0004	0.313 ± 0.0008
DLinear	None	0.381 ± 0.0478	0.421 ± 0.0199	0.245 ± 0.0004	0.298 ± 0.0009	0.166 ± 0.0000	0.264 ± 0.0002	0.434 ± 0.0000	0.295 ± 0.0000
	wDBA	0.386 ± 0.0602	0.422 ± 0.0311	0.245 ± 0.0031	0.296 ± 0.0062	0.173 ± 0.0000	0.263 ± 0.0004	0.434 ± 0.0000	0.295 ± 0.0000
	MBB	0.383 ± 0.0414	0.419 ± 0.0164	0.246 ± 0.0025	0.299 ± 0.0056	0.173 ± 0.0001	0.273 ± 0.0001	0.450 ± 0.0000	0.315 ± 0.0002
	RobustTAD-m/p	0.363 ± 0.0539	0.412 ± 0.0234	0.245 ± 0.0005	0.297 ± 0.0008	0.167 ± 0.0000	0.264 ± 0.0001	0.434 ± 0.0000	0.296 ± 0.0000
	FreqAdd	0.385 ± 0.0344	0.430 ± 0.0166	0.247 ± 0.0009	0.299 ± 0.0018	0.170 ± 0.0000	0.269 ± 0.0001	0.435 ± 0.0000	0.300 ± 0.0000
	FreqPool	0.286 ± 0.0127	0.377 ± 0.0081	0.261 ± 0.0009	0.316 ± 0.0017	0.182 ± 0.0000	0.280 ± 0.0002	0.466 ± 0.0000	0.332 ± 0.0001
	Upsample	0.242 ± 0.0082	0.351 ± 0.0067	0.246 ± 0.0006	0.298 ± 0.0010	0.217 ± 0.0004	0.310 ± 0.0005	0.519 ± 0.0007	0.397 ± 0.0011
	STAug	0.383 ± 0.0449	0.421 ± 0.0183	0.345 ± 0.0164	0.391 ± 0.0151	-	-	-	-
	Freq-Mask/Mix	0.317 ± 0.0146	0.393 ± 0.0066	0.245 ± 0.0008	0.296 ± 0.0017	0.167 ± 0.0000	0.265 ± 0.0002	0.436 ± 0.0000	0.298 ± 0.0000
	Wave-Mask/Mix	0.375 ± 0.0336	0.418 ± 0.0142	0.245 ± 0.0006	0.297 ± 0.0012	0.166 ± 0.0000	0.264 ± 0.0001	0.434 ± 0.0000	0.296 ± 0.0000
	Dominant Shuffle	0.340 ± 0.0525	0.408 ± 0.0220	0.246 ± 0.0003	0.297 ± 0.0007	0.167 ± 0.0000	0.265 ± 0.0002	0.435 ± 0.0001	0.297 ± 0.0003
	TPS (Ours)	0.237 ± 0.0035	0.349 ± 0.0026	0.239 ± 0.0003	0.285 ± 0.0007	0.166 ± 0.0000	0.263 ± 0.0002	0.432 ± 0.0000	0.292 ± 0.0000
PatchTST	None	0.381 ± 0.0082	0.413 ± 0.0052	0.237 ± 0.0008	0.272 ± 0.0007	0.163 ± 0.0003	0.255 ± 0.0003	0.411 ± 0.0005	0.274 ± 0.0005
	wDBA	0.391 ± 0.0165	0.418 ± 0.0078	0.230 ± 0.0005	0.266 ± 0.0003	0.162 ± 0.0004	0.255 ± 0.0004	0.412 ± 0.0005	0.274 ± 0.0004
	MBB	0.391 ± 0.0091	0.417 ± 0.0053	0.231 ± 0.0006	0.266 ± 0.0009	0.164 ± 0.0004	0.257 ± 0.0003	0.413 ± 0.0004	0.275 ± 0.0005
	RobustTAD-m/p	0.395 ± 0.0152	0.419 ± 0.0074	0.236 ± 0.0007	0.271 ± 0.0008	0.162 ± 0.0001	0.254 ± 0.0002	0.408 ± 0.0004	0.273 ± 0.0004
	FreqAdd	0.375 ± 0.0083	0.414 ± 0.0051	0.239 ± 0.0007	0.274 ± 0.0007	0.164 ± 0.0003	0.258 ± 0.0003	0.408 ± 0.0004	0.275 ± 0.0004
	FreqPool	0.401 ± 0.0136	0.424 ± 0.0063	0.238 ± 0.0007	0.273 ± 0.0009	0.163 ± 0.0005	0.255 ± 0.0003	0.412 ± 0.0006	0.278 ± 0.0010
	Upsample	0.361 ± 0.0064	0.415 ± 0.0039	0.234 ± 0.0007	0.270 ± 0.0010	0.165 ± 0.0006	0.257 ± 0.0006	0.415 ± 0.0007	0.281 ± 0.0009
	STAug	0.381 ± 0.0086	0.413 ± 0.0045	0.265 ± 0.0011	0.303 ± 0.0013	-	-	-	-
	Freq-Mask/Mix	0.410 ± 0.0083	0.428 ± 0.0050	0.238 ± 0.0005	0.273 ± 0.0007	0.164 ± 0.0001	0.256 ± 0.0002	0.412 ± 0.0003	0.276 ± 0.0003
	Wave-Mask/Mix	0.380 ± 0.0103	0.426 ± 0.0065	0.236 ± 0.0004	0.271 ± 0.0005	0.163 ± 0.0002	0.256 ± 0.0002	0.407 ± 0.0002	0.273 ± 0.0002
	Dominant Shuffle	0.367 ± 0.0082	0.418 ± 0.0036	0.241 ± 0.0006	0.275 ± 0.0051	0.163 ± 0.0002	0.255 ± 0.0001	0.409 ± 0.0004	0.274 ± 0.0002
	TPS (Ours)	0.346 ± 0.0028	0.397 ± 0.0019	0.230 ± 0.0005	0.266 ± 0.0009	0.162 ± 0.0001	0.254 ± 0.0002	0.408 ± 0.0004	0.274 ± 0.0004
TiDE	None	0.382 ± 0.0023	0.414 ± 0.0010	0.240 ± 0.0004	0.279 ± 0.0005	0.162 ± 0.0000	0.255 ± 0.0000	0.440 ± 0.0003	0.319 ± 0.0003
	wDBA	0.379 ± 0.0011	0.412 ± 0.0010	0.240 ± 0.0004	0.277 ± 0.0002	0.162 ± 0.0000	0.255 ± 0.0002	0.440 ± 0.0005	0.320 ± 0.0005
	MBB	0.380 ± 0.0029	0.413 ± 0.0008	0.240 ± 0.0004	0.279 ± 0.0003	0.165 ± 0.0003	0.259 ± 0.0004	0.442 ± 0.0005	0.321 ± 0.0003
	RobustTAD-m/p	0.385 ± 0.0023	0.415 ± 0.0003	0.240 ± 0.0004	0.279 ± 0.0005	0.162 ± 0.0000	0.255 ± 0.0001	0.440 ± 0.0005	0.319 ± 0.0004
	FreqAdd	0.381 ± 0.0015	0.415 ± 0.0010	0.242 ± 0.0004	0.282 ± 0.0005	0.165 ± 0.0000	0.259 ± 0.0000	0.436 ± 0.0003	0.314 ± 0.0004
	FreqPool	0.387 ± 0.0020	0.416 ± 0.0016	0.253 ± 0.0006	0.290 ± 0.0002	0.172 ± 0.0002	0.267 ± 0.0002	0.465 ± 0.0004	0.343 ± 0.0005
	Upsample	0.372 ± 0.0014	0.409 ± 0.0007	0.238 ± 0.0006	0.275 ± 0.0011	0.181 ± 0.0002	0.277 ± 0.0004	0.488 ± 0.0007	0.372 ± 0.0006
	STAug	0.383 ± 0.0017	0.414 ± 0.0013	0.341 ± 0.0213	0.338 ± 0.0106	-	-	-	-
	Freq-Mask/Mix	0.421 ± 0.0071	0.430 ± 0.0023	0.242 ± 0.0004	0.280 ± 0.0004	0.164 ± 0.0000	0.258 ± 0.0000	0.439 ± 0.0003	0.318 ± 0.0031
	Wave-Mask/Mix	0.382 ± 0.0029	0.413 ± 0.0011	0.241 ± 0.0004	0.280 ± 0.0004	0.162 ± 0.0000	0.255 ± 0.0001	0.438 ± 0.0003	0.318 ± 0.0003
	Dominant Shuffle	0.376 ± 0.0040	0.413 ± 0.0015	0.243 ± 0.0003	0.281 ± 0.0003	0.163 ± 0.0000	0.257 ± 0.0000	0.439 ± 0.0003	0.318 ± 0.0003
	TPS (Ours)	0.367 ± 0.0009	0.407 ± 0.0003	0.234 ± 0.0001	0.272 ± 0.0013	0.162 ± 0.0000	0.255 ± 0.0000	0.437 ± 0.0002	0.316 ± 0.0002
LightTS	None	0.416 ± 0.0348	0.462 ± 0.0133	0.236 ± 0.0027	0.290 ± 0.0036	0.182 ± 0.0088	0.288 ± 0.0106	0.445 ± 0.0053	0.328 ± 0.0054
	wDBA	0.414 ± 0.0214	0.453 ± 0.0110	0.235 ± 0.0031	0.287 ± 0.0030	0.183 ± 0.0077	0.290 ± 0.0088	0.445 ± 0.0042	0.327 ± 0.0034
	MBB	0.427 ± 0.0182	0.462 ± 0.0102	0.237 ± 0.0026	0.290 ± 0.0031	0.185 ± 0.0057	0.294 ± 0.0434	0.449 ± 0.0058	0.330 ± 0.0030
	RobustTAD-m/p	0.419 ± 0.0304	0.459 ± 0.0156	0.236 ± 0.0031	0.290 ± 0.0033	0.180 ± 0.0064	0.285 ± 0.0082	0.440 ± 0.0104	0.319 ± 0.0105
	FreqAdd	0.425 ± 0.0374	0.465 ± 0.0196	0.238 ± 0.0028	0.293 ± 0.0046	0.179 ± 0.0065	0.284 ± 0.0070	0.448 ± 0.0068	0.329 ± 0.0067
	FreqPool	0.326 ± 0.0977	0.396 ± 0.0075	0.239 ± 0.0019	0.288 ± 0.0015	0.186 ± 0.0072	0.290 ± 0.0075	0.468 ± 0.0087	0.347 ± 0.0060
	Upsample	0.289 ± 0.0128	0.394 ± 0.0080	0.235 ± 0.0028	0.285 ± 0.0018	0.187 ± 0.0058	0.294 ± 0.0072	0.474 ± 0.0102	0.354 ± 0.0088
	STAug	0.410 ± 0.0366	0.452 ± 0.0160	0.287 ± 0.0144	0.339 ± 0.0165	-	-	-	-
	Freq-Mask/Mix	0.372 ± 0.0360	0.440 ± 0.0096	0.235 ± 0.0020	0.287 ± 0.0026	0.179 ± 0.0062	0.282 ± 0.0061	0.441 ± 0.0027	0.322 ± 0.0023
	Wave-Mask/Mix	0.421 ± 0.0310	0.453 ± 0.0142	0.235 ± 0.0039	0.289 ± 0.0059	0.178 ± 0.0059	0.284 ± 0.0077	0.440 ± 0.0047	0.322 ± 0.0039
	Dominant Shuffle	0.358 ± 0.0138	0.432 ± 0.0089	0.235 ± 0.0032	0.287 ± 0.0034	0.192 ± 0.0111	0.296 ± 0.0093	0.445 ± 0.0051	0.327 ± 0.0055
	TPS (Ours)	0.273 ± 0.0183	0.385 ± 0.0118	0.228 ± 0.0026	0.277 ± 0.0019	0.174 ± 0.0030	0.278 ± 0.0050	0.439 ± 0.0071	0.320 ± 0.0055

Table 5: Short-term traffic forecasting with PatchTST on PeMS-{03,04,07,08}. For each prediction length, results are reported as mean±std over 5 runs. Best results are highlighted in bold, and second-best results are underlined.

	Method	12		24		36		48	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
PeMS03	None	0.084 ± 0.0024	0.200 ± 0.0046	0.115 ± 0.0054	0.238 ± 0.0115	0.127 ± 0.0037	0.239 ± 0.0038	0.147 ± 0.0072	0.259 ± 0.0101
	wDBA [7]	0.086 ± 0.0060	0.203 ± 0.0092	0.120 ± 0.0043	0.230 ± 0.0014	0.135 ± 0.0070	0.265 ± 0.0106	0.158 ± 0.0013	0.289 ± 0.0024
	MBB [3]	0.075 ± 0.0005	<u>0.183 ± 0.0008</u>	0.107 ± 0.0017	0.221 ± 0.0025	0.140 ± 0.0013	0.259 ± 0.0051	0.151 ± 0.0013	0.262 ± 0.0015
	RobustTAD-m/p [8]	0.090 ± 0.0016	0.206 ± 0.0041	0.118 ± 0.0041	0.240 ± 0.0062	0.128 ± 0.0065	0.247 ± 0.0089	0.145 ± 0.0036	0.257 ± 0.0020
	FreqAdd [12]	0.088 ± 0.0041	0.205 ± 0.0082	0.119 ± 0.0063	0.244 ± 0.0110	0.135 ± 0.0032	0.253 ± 0.0047	0.149 ± 0.0047	0.261 ± 0.0036
	FreqPool [5]	0.078 ± 0.0009	0.190 ± 0.0022	0.106 ± 0.0053	0.227 ± 0.0103	0.120 ± 0.0027	0.237 ± 0.0036	0.145 ± 0.0032	0.265 ± 0.0067
	Upsample [9]	0.080 ± 0.0024	0.195 ± 0.0051	0.110 ± 0.0066	0.235 ± 0.0115	<u>0.119 ± 0.0054</u>	0.236 ± 0.0069	0.140 ± 0.0061	0.256 ± 0.0079
	STAug [11]	<u>0.076 ± 0.0014</u>	0.185 ± 0.0032	<u>0.102 ± 0.0015</u>	<u>0.214 ± 0.0017</u>	0.125 ± 0.0027	0.238 ± 0.0038	0.147 ± 0.0029	0.260 ± 0.0051
	Freq-Mask/Mix [4]	0.084 ± 0.0030	0.198 ± 0.0058	0.123 ± 0.0067	0.253 ± 0.0142	0.134 ± 0.0077	0.251 ± 0.0090	0.153 ± 0.0082	0.268 ± 0.0114
	Wave-Mask/Mix [1]	0.085 ± 0.0080	0.196 ± 0.0079	0.106 ± 0.0029	0.221 ± 0.0044	0.122 ± 0.0064	<u>0.235 ± 0.0064</u>	0.145 ± 0.0061	0.258 ± 0.0070
	Dominant Shuffle [13]	0.087 ± 0.0009	0.199 ± 0.0039	0.111 ± 0.0034	0.235 ± 0.0062	0.122 ± 0.0021	0.238 ± 0.0032	<u>0.139 ± 0.0044</u>	<u>0.253 ± 0.0061</u>
	TPS (Ours)	0.076 ± 0.0024	0.181 ± 0.0019	0.094 ± 0.0017	0.207 ± 0.0029	0.114 ± 0.0023	0.230 ± 0.0039	0.133 ± 0.0056	0.245 ± 0.0105
PeMS04	None	0.094 ± 0.0007	0.207 ± 0.0008	0.121 ± 0.0013	0.238 ± 0.0021	0.155 ± 0.0070	0.272 ± 0.0101	0.171 ± 0.0089	0.280 ± 0.0067
	wDBA	0.093 ± 0.0019	0.205 ± 0.0044	0.125 ± 0.0015	0.241 ± 0.0022	0.150 ± 0.0033	0.259 ± 0.0036	0.169 ± 0.0018	0.277 ± 0.0049
	MBB	0.095 ± 0.0001	0.210 ± 0.0014	0.133 ± 0.0081	0.253 ± 0.0116	0.153 ± 0.0057	0.271 ± 0.0091	0.176 ± 0.0045	0.285 ± 0.0056
	RobustTAD-m/p	0.095 ± 0.0017	0.209 ± 0.0027	0.125 ± 0.0054	0.241 ± 0.0068	0.147 ± 0.0017	0.263 ± 0.0047	0.168 ± 0.0030	0.281 ± 0.0047
	FreqAdd	0.098 ± 0.0047	0.213 ± 0.0068	0.128 ± 0.0036	0.244 ± 0.0037	0.155 ± 0.0034	0.268 ± 0.0072	0.175 ± 0.0042	0.284 ± 0.0062
	FreqPool	0.093 ± 0.0015	<u>0.204 ± 0.0013</u>	0.119 ± 0.0014	<u>0.235 ± 0.0032</u>	<u>0.142 ± 0.0037</u>	0.260 ± 0.0056	0.157 ± 0.0036	0.271 ± 0.0031
	Upsample	0.096 ± 0.0020	0.210 ± 0.0044	0.127 ± 0.0013	0.247 ± 0.0028	0.149 ± 0.0039	0.272 ± 0.0083	0.171 ± 0.0065	0.285 ± 0.0055
	STAug	0.093 ± 0.0012	0.204 ± 0.0024	0.122 ± 0.0026	0.238 ± 0.0044	0.148 ± 0.0031	0.262 ± 0.0050	0.178 ± 0.0043	0.290 ± 0.0036
	Freq-Mask/Mix	0.101 ± 0.0005	0.218 ± 0.0018	0.133 ± 0.0042	0.246 ± 0.0043	0.158 ± 0.0071	0.272 ± 0.0070	0.178 ± 0.0084	0.292 ± 0.0077
	Wave-Mask/Mix	<u>0.092 ± 0.0008</u>	0.205 ± 0.0014	0.119 ± 0.0036	0.235 ± 0.0038	0.147 ± 0.0023	<u>0.259 ± 0.0026</u>	0.172 ± 0.0048	0.285 ± 0.0051
	Dominant Shuffle	0.097 ± 0.0015	0.213 ± 0.0034	0.123 ± 0.0026	0.240 ± 0.0017	0.148 ± 0.0056	0.267 ± 0.0066	0.169 ± 0.0018	0.287 ± 0.0041
	TPS (Ours)	0.091 ± 0.0012	0.202 ± 0.0018	0.112 ± 0.0023	0.227 ± 0.0029	0.133 ± 0.0039	0.250 ± 0.0046	0.163 ± 0.0063	0.274 ± 0.0068
PeMS07	None	0.073 ± 0.0011	0.188 ± 0.0034	0.101 ± 0.0061	0.222 ± 0.0100	0.129 ± 0.0050	0.252 ± 0.0115	0.164 ± 0.0059	0.300 ± 0.0081
	wDBA	0.075 ± 0.0022	0.190 ± 0.0042	0.100 ± 0.0007	0.224 ± 0.0053	0.120 ± 0.0120	0.245 ± 0.0168	0.142 ± 0.0050	0.273 ± 0.0038
	MBB	0.078 ± 0.0010	0.201 ± 0.0030	0.107 ± 0.0010	0.236 ± 0.0022	0.143 ± 0.0062	0.284 ± 0.0051	0.172 ± 0.0042	0.310 ± 0.0048
	RobustTAD-m/p	0.076 ± 0.0020	0.190 ± 0.0044	0.103 ± 0.0036	0.231 ± 0.0076	0.120 ± 0.0027	0.246 ± 0.0043	0.146 ± 0.0069	0.274 ± 0.0117
	FreqAdd	0.076 ± 0.0018	0.194 ± 0.0044	0.106 ± 0.0033	0.238 ± 0.0078	0.136 ± 0.0100	0.273 ± 0.0153	0.154 ± 0.0118	0.290 ± 0.0179
	FreqPool	0.076 ± 0.0095	0.187 ± 0.0116	<u>0.095 ± 0.0040</u>	0.219 ± 0.0093	0.124 ± 0.0164	0.253 ± 0.0273	0.137 ± 0.0099	0.268 ± 0.0161
	Upsample	0.078 ± 0.0060	0.193 ± 0.0100	0.104 ± 0.0043	0.228 ± 0.0079	0.121 ± 0.0076	0.248 ± 0.0124	<u>0.134 ± 0.0063</u>	0.263 ± 0.0099
	STAug	<u>0.071 ± 0.0020</u>	<u>0.181 ± 0.0060</u>	0.114 ± 0.0054	<u>0.218 ± 0.0045</u>	0.129 ± 0.0082	0.255 ± 0.0075	0.159 ± 0.0054	0.285 ± 0.0076
	Freq-Mask/Mix	0.079 ± 0.0037	0.194 ± 0.0060	0.110 ± 0.0074	0.239 ± 0.0130	0.126 ± 0.0047	0.249 ± 0.0051	0.137 ± 0.0075	<u>0.258 ± 0.0093</u>
	Wave-Mask/Mix	0.074 ± 0.0053	0.190 ± 0.0128	0.096 ± 0.0025	0.221 ± 0.0040	<u>0.118 ± 0.0037</u>	0.248 ± 0.0068	0.130 ± 0.0022	0.254 ± 0.0032
	Dominant Shuffle	0.078 ± 0.0031	0.195 ± 0.0080	0.100 ± 0.0063	0.222 ± 0.0113	0.118 ± 0.0076	0.244 ± 0.0124	0.139 ± 0.0095	0.264 ± 0.0156
	TPS (Ours)	0.070 ± 0.0023	0.179 ± 0.0038	0.090 ± 0.0030	0.207 ± 0.0045	0.118 ± 0.0113	0.244 ± 0.0203	0.143 ± 0.0075	0.270 ± 0.0120
PeMS08	None	0.099 ± 0.0056	0.206 ± 0.0080	0.143 ± 0.0034	0.243 ± 0.0061	0.186 ± 0.0050	0.285 ± 0.0117	0.206 ± 0.0140	0.302 ± 0.0123
	wDBA	0.102 ± 0.0069	0.210 ± 0.0128	0.146 ± 0.0045	0.239 ± 0.0053	0.200 ± 0.0035	0.282 ± 0.0198	0.232 ± 0.0124	0.316 ± 0.0175
	MBB	0.095 ± 0.0006	0.203 ± 0.0012	0.149 ± 0.0044	0.245 ± 0.0092	0.188 ± 0.0031	0.269 ± 0.0015	0.244 ± 0.0074	0.324 ± 0.0161
	RobustTAD-m/p	0.101 ± 0.0088	0.210 ± 0.0099	0.145 ± 0.0036	0.242 ± 0.0035	0.182 ± 0.0052	0.276 ± 0.0124	0.208 ± 0.0155	0.308 ± 0.0128
	FreqAdd	0.103 ± 0.0058	0.217 ± 0.0082	0.151 ± 0.0069	0.255 ± 0.0089	0.185 ± 0.0072	0.296 ± 0.0076	0.206 ± 0.0100	0.304 ± 0.0108
	FreqPool	0.097 ± 0.0031	0.203 ± 0.0044	<u>0.128 ± 0.0093</u>	0.238 ± 0.0096	0.161 ± 0.0028	0.270 ± 0.0130	<u>0.187 ± 0.0076</u>	0.296 ± 0.0096
	Upsample	0.093 ± 0.0039	0.205 ± 0.0053	0.129 ± 0.0036	0.240 ± 0.0034	<u>0.160 ± 0.0110</u>	<u>0.267 ± 0.0119</u>	0.181 ± 0.0084	0.279 ± 0.0093
	STAug	0.094 ± 0.0050	0.205 ± 0.0089	0.131 ± 0.0042	<u>0.236 ± 0.0051</u>	0.167 ± 0.0070	0.268 ± 0.0114	0.206 ± 0.0077	<u>0.295 ± 0.0093</u>
	Freq-Mask/Mix	0.101 ± 0.0014	0.206 ± 0.0022	0.150 ± 0.0051	0.252 ± 0.0050	0.194 ± 0.0102	0.302 ± 0.0139	0.221 ± 0.0112	0.304 ± 0.0187
	Wave-Mask/Mix	<u>0.092 ± 0.0024</u>	<u>0.202 ± 0.0046</u>	0.130 ± 0.0077	0.248 ± 0.0089	0.167 ± 0.0149	0.276 ± 0.0156	0.206 ± 0.0272	0.306 ± 0.0310
	Dominant Shuffle	0.096 ± 0.0031	0.205 ± 0.0041	0.141 ± 0.0072	0.244 ± 0.0074	0.178 ± 0.0088	0.280 ± 0.0091	0.210 ± 0.0043	0.293 ± 0.0052
	TPS (Ours)	0.089 ± 0.0022	0.197 ± 0.0036	0.113 ± 0.0033	0.223 ± 0.0073	0.139 ± 0.0047	0.246 ± 0.0050	0.198 ± 0.0104	0.295 ± 0.0097

Table 6: Component-wise ablation results of TPS on ETT-{h1,h2,m1,m2} using DLinear. Values are reported as mean ± std MSE/MAE over five runs, averaged across prediction lengths {96, 192, 336, 720}. Best results are highlighted in bold, and second-best results are underlined.

Methods	ETTh1		ETTh2		ETTm1		ETTm2	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
None	0.438 ± 0.0194	0.449 ± 0.0162	0.464 ± 0.0099	0.462 ± 0.0053	0.361 ± 0.0010	0.383 ± 0.0015	0.276 ± 0.0099	0.339 ± 0.0093
TPS	0.410 ± 0.0036	0.425 ± 0.0031	0.369 ± 0.0056	0.408 ± 0.0039	0.354 ± 0.0006	0.377 ± 0.0006	0.261 ± 0.0018	0.324 ± 0.0027
- Variance Score	0.417 ± 0.0166	0.430 ± 0.0129	<u>0.370 ± 0.0066</u>	<u>0.409 ± 0.0048</u>	<u>0.355 ± 0.0006</u>	0.377 ± 0.0005	0.261 ± 0.0024	0.324 ± 0.0033
- Temporal Patching	<u>0.416 ± 0.0083</u>	<u>0.430 ± 0.0040</u>	0.379 ± 0.0092	0.424 ± 0.0057	0.376 ± 0.0018	0.397 ± 0.0009	<u>0.267 ± 0.0031</u>	<u>0.332 ± 0.0039</u>
- Data-Label Coherence	0.443 ± 0.0158	0.451 ± 0.0144	0.438 ± 0.0146	0.447 ± 0.0074	0.364 ± 0.0018	0.386 ± 0.0022	0.290 ± 0.0122	0.352 ± 0.0103
+ Frequency Domain	0.437 ± 0.0096	0.448 ± 0.0077	0.470 ± 0.0267	0.464 ± 0.0132	0.363 ± 0.0011	0.384 ± 0.0016	0.285 ± 0.0082	0.345 ± 0.0065

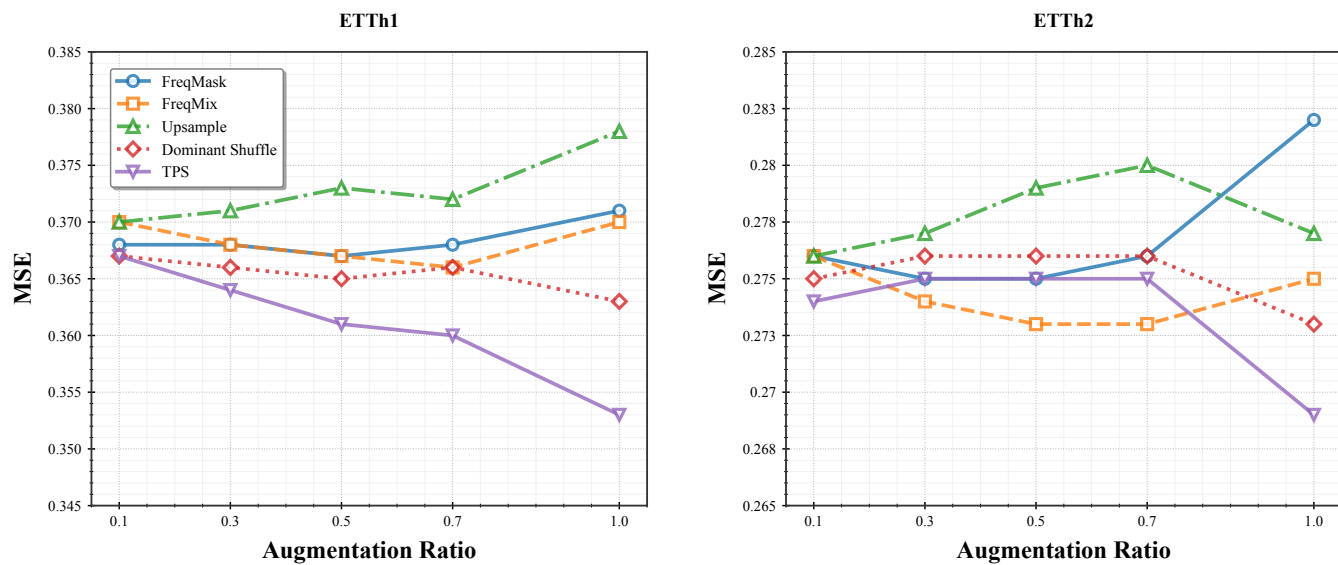


Figure 2: Effect of varying augmentation ratios (0.1 to 1.0) on the MSE performance of different augmentation methods using the PatchTST model with prediction length 96 on the ETTh1 and ETTh2 datasets. The augmentation ratio indicates the proportion of augmented samples used during training.

Table 7: Summary of selected UCR univariate time series classification datasets used in our experiments. The table includes dataset category, name, training and test sizes, input sequence lengths, and numbers of classes [6].

Type	Dataset	Train	Test	Length	Classes
Device	ACSF1	100	100	1460	10
	HouseTwenty	34	101	3000	2
	ScreenType	375	375	720	3
ECG	ECG5000	500	4500	140	5
	ECG200	100	100	96	2
	ECGFiveDays	23	861	136	2
Image	Adiac	390	391	176	37
	FaceFour	24	88	350	4
	FaceAll	560	1690	131	14
	HandOutlines	1000	370	2709	2
	MiddlePhalanxTW	399	154	80	6
	PhalangesOutlinesCorrect	1800	858	80	2
	ShapesAll	600	600	512	60
Motion	Haptics	155	308	1092	5
	WormsTwoClass	181	77	900	2
	InlineSkate	100	550	1882	7
Sensor	Car	60	60	577	4
	Earthquakes	322	139	512	2
	FordA	3601	1320	500	2
	FordB	3636	810	500	2
	ItalyPowerDemand	67	1029	24	2
	Lightning2	60	61	637	2
	StarLightCurves	1000	8236	1024	3
Spectro	Beef	30	30	470	5
	EthanolLevel	504	500	1751	4
	Wine	57	54	234	2
	Meat	60	60	448	3
	OliveOil	30	30	570	4
Simulated/Audio	ChlorineConcentration	467	3840	166	3
	Phoneme	214	1896	1024	39

Table 8: Summary of selected UEA multivariate time series classification datasets used in our experiments. The table includes training and test sizes, input dimensionality, sequence length, and number of classes [2].

Dataset	Train	Test	Dim.	Length	Classes
AtrialFibrillation	15	15	2	640	3
Cricket	108	72	6	1197	12
DuckDuckGeese	60	40	1345	270	5
ERing	30	30	4	65	6
EthanolConcentration	261	263	3	1751	4
LSST	2459	2466	6	36	14
Libras	180	180	2	45	15
FaceDetection	5890	3524	144	62	2
FingerMovements	316	100	28	50	2
MotorImagery	278	100	64	3000	2

References

- [1] Dona Arabi, Jafar Bakhshaliyev, Ayse Coskuner, Kiran Madhusudhanan, and Kami Serdar Uckardes. 2024. Wave-Mask/Mix: Exploring Wavelet-Based Augmentations for Time Series Forecasting. [arXiv:2408.10951](#)
- [2] Anthony Bagnall, Hoang Anh Dau, Jason Lines, Michael Flynn, James Large, Aaron Bostrom, Paul Southam, and Eamonn Keogh. 2018. The UEA multivariate time series classification archive, 2018. [arXiv:1811.00075](#)
- [3] Kasun Bandara, Hansika Hewamalage, Yuan-Hao Liu, Yanfei Kang, and Christoph Bergmeir. 2021. Improving the accuracy of global forecasting models using time series data augmentation. *Pattern Recognition* 120 (2021), 108148.
- [4] Muxi Chen, Zhijian Xu, Ailing Zeng, and Qiang Xu. 2023. FrAug: Frequency Domain Augmentation for Time Series Forecasting. [arXiv:2302.09292](#)
- [5] Xi Chen, Cheng Ge, Ming Wang, and Jin Wang. 2023. Supervised contrastive few-shot learning for high-frequency time series. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 7069–7077.
- [6] Hoang Anh Dau, Eamonn Keogh, Kaveh Kamgar, Chin-Chia Michael Yeh, Yan Zhu, Shaghayegh Gharghabi, Chotirat Ann Ratanamahatana, Yanping, Bing Hu, Nurjahan Begum, Anthony Bagnall, Abdullah Mueen, Gustavo Batista, and Hexagon-ML. 2018. The UCR Time Series Classification Archive. https://www.cs.ucr.edu/~eamonn/time_series_data_2018.
- [7] Germain Forestier, François Petitjean, Hoang Anh Dau, Geoffrey I Webb, and Eamonn Keogh. 2017. Generating synthetic time series to augment sparse datasets. In *2017 IEEE international conference on data mining (ICDM)*. IEEE, 865–870.
- [8] Jingkun Gao, Xiaomin Song, Qingsong Wen, Pichao Wang, Liang Sun, and Huan Xu. 2021. RobustTAD: Robust Time Series Anomaly Detection via Decomposition and Convolutional Neural Networks. [arXiv:2002.09545](#)
- [9] Artemios-Anargyros Semenoglou, Evangelos Spiliotis, and Vassilios Assimakopoulos. 2023. Data augmentation for univariate time series forecasting with neural networks. *Pattern Recognition* 134 (2023), 109132.
- [10] Longhui Wei, An Xiao, Lingxi Xie, Xin Chen, Xiaopeng Zhang, and Qi Tian. 2020. Circumventing Outliers of AutoAugment with Knowledge Distillation. [arXiv:2003.11342](#)
- [11] Xiyuan Zhang, Ranak Roy Chowdhury, Jingbo Shang, Rajesh Gupta, and Dezhi Hong. 2023. Towards Diverse and Coherent Augmentation for Time-Series Forecasting. [arXiv:2303.14254](#)
- [12] Xiang Zhang, Ziyuan Zhao, Theodoros Tsiligkaridis, and Marinka Zitnik. 2022. Self-Supervised Contrastive Pre-Training For Time Series via Time-Frequency Consistency. [arXiv:2206.08496](#)
- [13] Kai Zhao, Zuojie He, Alex Hung, and Dan Zeng. 2024. Dominant Shuffle: A Simple Yet Powerful Data Augmentation for Time-series Prediction. [arXiv:2405.16456](#)